

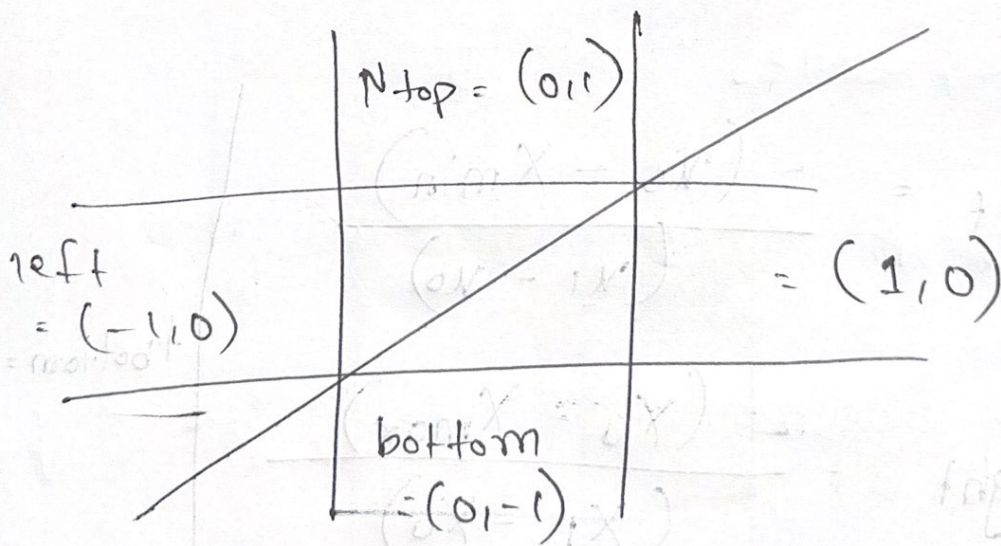
Parametric equation, $\vec{D}$

$$P(t) = P_0 + t \cdot (P_1 - P_0)$$

$$= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0)$$

$$= (x_0 + t(x_1 - x_0), y_0 + t(y_1 - y_0))$$

So, all points between $0 \leq t \leq 1$ are inside the line segment.



Cyons Beck

$(0,0), (300, 200)$

$$x_{\min} = 0$$

$$y_{\min} = 0$$

$$x_{\max} = 300$$

$$y_{\max} = 200.$$

(P) $(50, -125)$ to $(-100, 225)$.

$$x_0 = 50$$

$$x_1 = -100$$

$$y_0 = -125$$

$$y_1 = 225.$$

$$t_{\text{left}} = \frac{-(x_0 - x_{\min})}{(x_1 - x_0)}$$

$$t_{\text{right}} = \frac{-(x_0 - x_{\max})}{(x_1 - x_0)}$$

$$t_{\text{top}} = \frac{-(y_0 - y_{\max})}{(y_1 - y_0)}$$

$$t_{\text{bottom}} = \frac{-(y_0 - y_{\min})}{(y_1 - y_0)}$$



$(-1, 0)$ $(-150, 350)$
 $(150, 0)$

first we find $D_i = (x_1 - x_0, y_1 - y_0)$

for $D_i = (-150, 350)$.

if, $t_E > t_L$, line segment is outside.

$t_E < t_L$, line segment needs clipping.

initially, $t_E = 0$, $t_L = 1$.

Point entering

Point leaving

Boundary	No	No. D (+) → PL (-) → PE	t	PE/PL	t_E	t_L
left	$(-1, 0)$	150	$\frac{-(50-0)}{-100-50}$ $= 0.33$	PL	0	0.33 (निर्वात)
right	$(1, 0)$	-150	$\frac{-(50-300)}{-100-50}$ $= -1.67$	PE	0	0.33 (निर्वात)
bottom	$(0, -1)$	-350	$\frac{-(-125-0)}{250-(-125)}$ $= 0.357$	PE	0.357	0.33
top	$(0, 1)$	350	$\frac{-(-125-200)}{250-(-125)}$ $= 0.93$	PL	0.357	0.33

$\therefore t_E > t_L$

$$m = \frac{-(y_0 - y_{min})}{(y_1 - y_0)}$$

(Clipping ବ୍ୟାସ ଆକାର, Parametric Equation) ଏହା t and u ଏହା t ଏବଂ u ବ୍ୟାସ and u ବ୍ୟାସ t last :
 updated u ବ୍ୟାସ use $t < u$ (7)

Advantage

↳ Work in polygon

↳ Works in 3D

Problem

↳ Doesn't work in concave Polygon.